

Roberto García Antonio

Contact information

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Education

2023–2025 M.Sc., Mathematics, Universidad Veracruzana

Thesis Title: Reproductive Kernel Hilbert Spaces and Spherical Representations of Groups $SO(n)$ y $U(n)$

Supervisors: Dr. Josué Ramírez Ortega, Dra. Yessica Hernández Eliseo

Abstract: Proved the irreducibility of a representation on a RKHS using a criterion based on the existence of a fixed vector under the action of a closed group.

2019–2023 B.Sc., Mathematics, Universidad Veracruzana

Thesis Title: Álgebras de dimensión finita (finite-dimensional algebras)

Supervisors: Dr. Josué Ramírez Ortega, Dra. Yessica Hernández Eliseo

Abstract: Provided an explicit proof of the Jordan decomposition theorem to characterize the algebra generated by an operator on a finite-dimensional vector space.

Appointments

2023–2025 Teaching Assistant, Universidad Veracruzana

2023–2023 Undergraduate teacher, Universidad Veracruzana

2023–2023 Research Assistant, Dr. Josué Ramírez Ortega SNI 2, Area- Operator algebra

2024-present Analysis seminar, organizer Universidad Veracruzana

Teaching

2023 Topology, Universidad Veracruzana, Winter semester

Teaching Assistant

2024	Differential calculus on several variables	Universidad Veracruzana
2024	Integral calculus on several variables	Universidad Veracruzana
2024	Linear algebra	Universidad Veracruzana
2025	Group theory	Universidad Veracruzana

Talks and Conference Presentations

- [1] **Third Colloquium on Lie Theory and Applications, Faculty of Mathematics, Universidad Veracruzana, Mexico- February 2026**

Area: Harmonic analysis

Mini course: [Symmetry and analysis: From Lie groups to spherical harmonics](#)

- [2] **Speaker | Course on Harmonic Analysis and Toeplitz Operators CIMAT Mérida Aug 2025 to Jan 2026**

Delivered a series of 3 lectures covering Von Neumann Algebras, Harmonic analysis and Toeplitz Operators as part of the course curriculum.

- [3] **58th Congress of the Mexican Mathematical Society, Faculty of Sciences, BUAP, Puebla-October 2025**

Area: Analysis

Talk: [When linear algebra becomes analysis: a look at spectral theorems](#)

- [4] **UADY-CIMAT 2025 Postgraduate Meeting in Mathematics Faculty of Mathematics, UADY**
talk: [Teoría de representaciones y espacios de Hilbert con Núcleo reproductor \(Representation Theory and Reproducing Kernel Hilber spaces\)](#)

- [5] **Speaker | Course on C* and Toeplitz Operators CIMAT Mérida – Aug 2025 to Jan 2026**

Presented 4 lectures covering C*-algebras and their applications within the course structure

- [6] **Seminario de Análisis Matemático y Aplicaciones, Universidad Veracruzana, Facultad de Matemáticas, México**

Area: Analysis

talk: Pensando en integración-Integral de Pettis (Thinking About Integration – The Pettis Integral)

- [7] **Segundo coloquio de teoría de Lie y aplicaciones, Facultad de Matemáticas, Universidad Veracruzana, México**

Area: Lie Theory

talk: [Grupos de Lie y representaciones irreducibles: el grupo \$SO\(n\)\$ \(Lie groups and irreducible representations: \$SO\(n\)\$ group\).](#)

- [8] **Congreso de la sociedad mexicana de matemáticas, Facultad de Ciencias Exactas, Universidad Juárez del Estado de Durango (UJED), México**

Area: Analysis

talk: Representaciones irreducibles del grupo $SO(n)$ (irreducible representation of the $SO(n)$ group).

- [9] **Primer seminario de análisis matemático y aplicaciones UV, Facultad de Matemáticas, Universidad Veracruzana, México**

Area: Analysis

talk: Grupos topológicos con vistas a teoría de representaciones (Topological Groups from the Perspective of Representation Theory).

- [10] **Primer coloquio de teoría de Lie y aplicaciones, Facultad de Matemáticas, Universidad Veracruzana, México**

Area: Lie Theory

talk: [Formas bilineales y cuadráticas \(Bilinear and quadratics forms\).](#)

- [11] **Escuela de verano 2023 CIMAT, CIMAT-Unidad Merida, Mexico**

Area: Operator Theory

Talk: Algebras y una prueba del Teorema de descomposición de Jordan (Algebras and a Proof of the Jordan Decomposition Theorem).

Schools and workshops

- [1] **School of differential geometry and topology**, CIMAT, Guanajuato, 2025.
- [2] **Third school of geometric group theory**, UNAM-CCM, Morelia, 2024.
- [3] **Cimat summer school**, CIMAT, Merida, 2023.
- [4] **Eighth Mexican Congress on Fixed Points and Applications**, CIMAT, Merida, 2023.
- [5] **International Workshop on Operator Theory on Function Spaces**, UNIVERSIDAD VERACRUZANA-CINVESTAV, Veracruz, 2022.

Professional Activities

Science communication

- **Workshop for mathematics education-Colegio las Américas, Xalapa, 2023**

Workshop

Codificando y decodificando imágenes

- **XVI Xalapa Physics Meeting, 2019**

Workshop

¿Qué onda con las ondas?, Grupos LUX

Organizations

Some academic activities that I have co-organized

- [Colloquium on Topology, Geometry and Applications](#)
- [2nd Colloquium on Lie Theory and Applications](#)
- [6th Alumni Forum and Winter Mathematics Conference of the Faculty of Mathematics](#)
- [Seminar on mathematical analysis and applications at the University of Veracruz.](#)

In 2024, I proposed the initiative to hold a seminar on mathematical analysis at the University of Veracruz. Together with university professors and colleagues, we have been organizing this project for two years.

Languages

- Spanish-Native
- English-Advanced, B2 certificated by **TOEFEL IBT**
- German-Beginner

References

Available on request.